Predicting Natural Gas Volatility Across Regimes: Structural Trends, Asymmetric Semivariance, and a Sub-Second Tensor Pipeline

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Abstract

Natural gas (NG) futures exhibit pronounced multi-year price cycles interrupted by acute, highly localized volatility explosions. In this paper, I investigate whether Markov-Switching models (the Hamilton filter) can effectively detrend cyclical variance and forecast out-of-sample realized volatility. Evaluating 5-minute realized variance and semivariance measures obtained from the VOLatility Archive for Realized Estimates (VOLARE) database spanning 2009 to 2026, I benchmark 36 model configurations spanning pure realized variance (RV), Heterogeneous Autoregressive (HAR) dynamics, and Realized Semivariance (RS) with up to seven latent regimes, estimated across both linear and logarithmic scales. Because Jensen's inequality biases naive log-variance transformations, all models are evaluated on the true variance scale using the robust, asymmetric QLIKE loss function. Empirically, a 5-regime log-HAR specification achieves the global minimum QLIKE loss (0.2131). In parallel, I present a high-performance quantitative engineering framework: by vectorizing expanding-window ordinary least squares into 3D tensors using batched operations in PyTorch, the baseline linear OLS models are solved across all 875 expanding windows simultaneously in under 10 milliseconds (in contrast to iterative loops in Python/statsmodels which require several minutes on multi-model pipelines). Furthermore, by JIT-compiling the Hamilton log-likelihood filter via Numba, the entire 18-model non-linear regime comparison up to 7 regimes completes in just 18 minutes—a workflow that would take over 87 hours (3.6 days) using standard statistical libraries like `statsmodels`.

Keywords: Natural Gas, Volatility Forecasting, VOLARE, Hamilton Filter, Markov-Switching HAR, Realized Semivariance, QLIKE, High-Performance Python, PyTorch, Numba.

1 Introduction and Empirical Motivation

Energy derivatives, and Henry Hub Natural Gas (NG) futures in particular, represent one of the most volatile asset classes in global financial markets. Physical constraints—including finite pipeline capacity, geographical delivery bottlenecks, weather-dependent consumption cycles (winter heating and summer cooling), and inelastic near-term supply—make the commodity prone to severe structural regimes.

Figure 1 illustrates this fundamental empirical duality over an extensive seventeen-year sample (October 2009 to September 2026):

1. **Price Trends vs. Volatility Spikes:** As shown in Panel (a), settlement prices exhibit persistent multi-year trends. I observe the post-2009 shale boom depression where prices settled between \$2.00 and \$4.00/MMBtu, followed by sudden geopolitical and meteorological dislocations pushing prices toward \$10.00/MMBtu in 2022.



Figure 1: **Henry Hub Natural Gas Futures Market Dynamics (2009–2026).** Panel (a) illustrates the daily settlement price in \$/MMBtu, highlighting multi-year macro trends and severe price spikes (such as the 2021–2022 energy crisis). Panel (b) depicts the 5-minute daily Realized Variance (RV_t), demonstrating severe volatility clustering and extreme, localized jump spikes (e.g., Winter Storm Uri in February 2021 where RV_t spiked past 0.065).

2. **Variance Intermittency:** Conversely, as shown in Panel (b), the 5-minute daily Realized Variance (RV_t) does not follow smooth, linear drifts. Instead, it is characterized by baseline quiescence interrupted by explosive, short-lived variance shocks—most notably during the February 2021 North American cold wave (Winter Storm Uri), where single-day realized variance exceeded 0.065, representing an annualized volatility well above 400%.

Traditional linear time-series models—such as linear Autoregressive Conditional Heteroskedasticity (GARCH) or standard linear Ordinary Least Squares (OLS) on past realized variance—assume stationary parameter vectors over the entire estimation window. When confronted with the abrupt regime shifts visible in Figure 1, static parameterizations fail: they can systematically overpredict volatility following a sudden spike and underpredict during calm consolidation periods.

This paper is structured around two central contributions:

- **Econometric Formulation:** I examine whether an M-regime Markov-Switching Hamilton filter, coupled with the Heterogeneous Autoregressive (HAR) framework and Realized Semivariance decomposition, can dynamically separate trend from transitory shocks, producing superior out-of-sample volatility forecasts under asymmetric loss (QLIKE).

- **Quantitative Engineering:** Expanding-window walk-forward evaluations are notoriously slow. In single-threaded statistical libraries (such as `statsmodels.tsa.regime_switching`), calculating a single out-of-sample forecast for a 2-regime model required approximately 20 seconds. Scaling this across an 875-day test window implies nearly 5 hours for a single model and over 87 hours ($\approx$ 3.6 days) for an 18-model suite up to 7 regimes. Furthermore, benchmark iterative OLS loops in standard Python/statsmodels pipelines previously took upwards of 4 minutes on multi-model batches. In this study, I evaluate 3 baseline linear OLS specifications and design a high-throughput pipeline utilizing 3D tensor-batched linear

algebra in PyTorch alongside a JIT-compiled Numba C-extension kernel for the Hamilton filter. This collapses linear OLS evaluations across all 875 expanding windows to under 10 milliseconds and compresses the full 18-model non-linear regime comparison down to just 18 minutes.

2 Data and High-Frequency Metrics

2.1 Realized Variance and Semivariance Decomposition

My empirical analysis uses high-frequency realized volatility measures obtained from the VOLARE (VOLatility Archive for Realized Estimates) repository (`https://volare.unime.it`), an open-access research infrastructure providing standardized realized estimators constructed from ultra-high-frequency tick-level data [3]. From the VOLARE commodity futures library, I retrieve daily 5-minute realized variance (RV_t) and signed semivariance (RS_t^+, RS_t^-) series for Henry Hub Natural Gas futures (`NG`) covering 4,393 raw trading days (yielding an effective sample of 4,371 trading days after trimming the initial 22-day monthly rolling window).

Let $p_{t,i}$ denote the log-settlement price of the Henry Hub front-month futures contract at 5-minute intraday interval i on trading day t, where $i = 1, \ldots, N$. The i-th intraday return is defined as $r_{t,i} = p_{t,i} - p_{t,i-1}$. Following Andersen and Bollerslev (1998), the daily 5-minute Realized Variance (RV_t) is given by:

$$RV_t = \sum_{i=1}^{N} r_{t,i}^2 \tag{1}$$

To capture the asymmetric impact of market direction (the leverage effect), I follow Barndorff-Nielsen, Kinnebrock, and Shephard (2010) and decompose total variance into continuous upside (RS_t^+) and downside (RS_t^-) Realized Semivariances:

$$RS_t^+ = \sum_{i=1}^{N} r_{t,i}^2 \mathbb{I}_{\{r_{t,i}>0\}}, \quad RS_t^- = \sum_{i=1}^{N} r_{t,i}^2 \mathbb{I}_{\{r_{t,i}<0\}} \tag{2}$$

where $\mathbb{I}_{\{.\}}$ is an indicator function. By definition, $RV_t = RS_t^+ + RS_t^-$. In commodity markets, supply squeeze events often generate asymmetric positive return jumps, unlike equity markets where negative jumps predominantly drive variance spikes.

2.2 Heterogeneous Autoregressive (HAR) Features

Following Corsi (2009), I capture the long-memory-like behavior of volatility without resorting to fractional integration by aggregating returns over multiple time horizons corresponding to daily (d), weekly (w), and monthly (m) trader horizons:

$$RV_t^{(w)} = \frac{1}{5} \sum_{k=0}^{4} RV_{t-k}, \quad RV_t^{(m)} = \frac{1}{22} \sum_{k=0}^{21} RV_{t-k} \tag{3}$$

An identical rolling construction is applied to RS_t^+ and RS_t^- to generate multi-period semivariance predictors. To strictly avoid look-ahead bias, all features are lagged by one trading day prior to target matching ($y_t = RV_t$).

3 Econometric Framework

3.1 Linear Benchmarks (OLS)

I establish three benchmark linear specifications:

1. **AR(1) Baseline (RV_5 Only):** Regresses tomorrow's variance solely on today's realized variance:
$$RV_t = \beta_0 + \beta_d RV_{t-1} + \varepsilon_t \tag{4}$$

2. **Corsi HAR (RV_5 HAR):** Incorporates daily, weekly, and monthly cascade components:
$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w RV_{t-1}^{(w)} + \beta_m RV_{t-1}^{(m)} + \varepsilon_t \tag{5}$$

3. **Semivariance HAR (RS HAR):** Decomposes each term into signed semivariances:
$$RV_t = \beta_0 + \beta_d^+ RS_{t-1}^+ + \beta_d^- RS_{t-1}^- + \beta_w^+ RS_{t-1}^{+(w)} + \beta_w^- RS_{t-1}^{-(w)} + \beta_m^+ RS_{t-1}^{+(m)} + \beta_m^- RS_{t-1}^{-(m)} + \varepsilon_t \tag{6}$$

3.2 Markov-Switching Regime Shift (Hamilton Filter)

To capture the cyclical transitions between quiescent states and extreme winter demand surges, parameters are allowed to depend on a latent discrete regime variable $S_t \in \{0, 1, \ldots, M-1\}$, governed by an $M \times M$ transition probability matrix P:

$$P_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i), \quad \sum_{j=0}^{M-1} P_{ij} = 1 \tag{7}$$

Conditional on regime $S_t = m$, the dependent variable follows:

$$y_t = X_t \beta_m + \varepsilon_{t,m}, \quad \varepsilon_{t,m} \sim \mathcal{N}(0, \sigma_m^2) \tag{8}$$

where X_t denotes the row vector of exogenous regressors and β_m is the regime-specific coefficient vector.

The Hamilton (1989) recursive filter computes the posterior state probabilities $\xi_t = [\mathbb{P}(S_t = 0 \mid \mathcal{F}_t), \ldots, \mathbb{P}(S_t = M-1 \mid \mathcal{F}_t)]^T$:

1. **State Prediction:** $\xi_{t|t-1} = P^T \xi_{t-1}$

2. **Conditional Density Calculation:** For each regime m,
$$\eta_t(m) = \frac{1}{\sqrt{2\pi}\sigma_m} \exp\left(-\frac{(y_t - X_t \beta_m)^2}{2\sigma_m^2} \right) \tag{9}$$

3. **Mixture Density & Log-Likelihood Update:**
$$f(y_t \mid \mathcal{F}_{t-1}) = \xi_{t|t-1}^T \eta_t, \quad \ln L = \sum_{t=1}^{T} \ln f(y_t \mid \mathcal{F}_{t-1}) \tag{10}$$

4. **Posterior State Inference:**
$$\xi_t = \frac{\xi_{t|t-1} \odot \eta_t}{f(y_t \mid \mathcal{F}_{t-1})} \tag{11}$$

The unconditional stationary probability vector $\xi_0 = \pi$ satisfies $(P^T - I + \mathbf{1}\mathbf{1}^T)^{-1}\mathbf{1}$. One-step-ahead out-of-sample forecasts are given by the expectation of the mixture:

$$\widehat{y}_{t+1|t} = \xi_{t+1|t}^T \mu_{t+1} = (P^T \xi_t)^T (X_{t+1} B) \tag{12}$$

where $B = [\beta_0, \beta_1, \ldots, \beta_{M-1}]$.

3.3 Evaluation, Transformations, and Jensen's Inequality

Estimating models in log-space ($\ln RV_t$) guarantees non-negativity in out-of-sample forecasts. However, by Jensen's inequality:

$$\mathbb{E}[RV_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\exp(\ln RV_t) \mid \mathcal{F}_{t-1}] > \exp(\mathbb{E}[\ln RV_t \mid \mathcal{F}_{t-1}]) \tag{13}$$

A naive exponentiation of log-predictions produces a structural under-prediction bias.

Furthermore, standard symmetric loss metrics such as Mean Squared Error (RMSE) and Mean Absolute Error (MAE) penalize over- and under-predictions equally on the variance scale. In financial risk management and derivative pricing, underestimating variance carries far more catastrophic tail risk than overestimation. Therefore, I evaluate all models on the true level scale using Patton's (2011) strictly consistent and scale-invariant Quasi-Likelihood (QLIKE) loss:

$$\text{QLIKE}(RV_t, \widehat{RV}_t) = \frac{RV_t}{\widehat{RV}_t} - \ln\left(\frac{RV_t}{\widehat{RV}_t}\right) - 1 \tag{14}$$

QLIKE is uniquely immune to the choice of variance proxy noise and heavily penalizes under-forecasting regimes.

4 High-Performance Quantitative Engineering

An expanding-window walk-forward evaluation over an out-of-sample horizon of $N_{\text{OOS}} = 875$ trading days requires 875 separate model re-calibrations per specification. In standard quantitative workflows relying on Python loops and conventional packages like `statsmodels`, this creates an intractable computational bottleneck:

1. **Iterative Linear OLS Latency:** While standard iterative loops in Python/statsmodels can take multiple minutes when running expanding-window regressions across larger model batches, sequential execution even for 3 baseline OLS models across 875 windows (2,625 separate matrix solves) introduces unnecessary latency.

2. **Markov-Switching Filter Complexity:** Non-linear numerical optimization is substantially more expensive. A single 1-step forecast window for a 2-regime specification in `statsmodels.MarkovRegression` takes approximately **20 seconds**. Evaluating an 875-day out-of-sample trajectory for just one model would therefore require:

$$875 \text{ windows} \times 20 \text{ s} = 17,500 \text{ s} \approx 4.86 \text{ hours} \tag{15}$$

   Benchmarking an entire suite of 18 regime-switching models (3 feature sets $\times$ 6 regime configurations $M \in \{2, 3, 4, 5, 6, 7\}$) in `statsmodels` would require:

$$18 \times 17,500 \text{ s} = 315,000 \text{ s} \approx 87.5 \text{ hours } (\approx 3.65 \text{ days}) \tag{16}$$

   of uninterrupted execution, not accounting for the quadratic growth in parameter space ($M^2 + MK + M$) and slower convergence at higher regimes ($M \geq 5$).

To eliminate these bottlenecks, I engineered a dual-engine architecture that accelerates linear backtests by over $20,000\times$ and compresses the multi-day non-linear regime benchmarking suite to **18 minutes**.

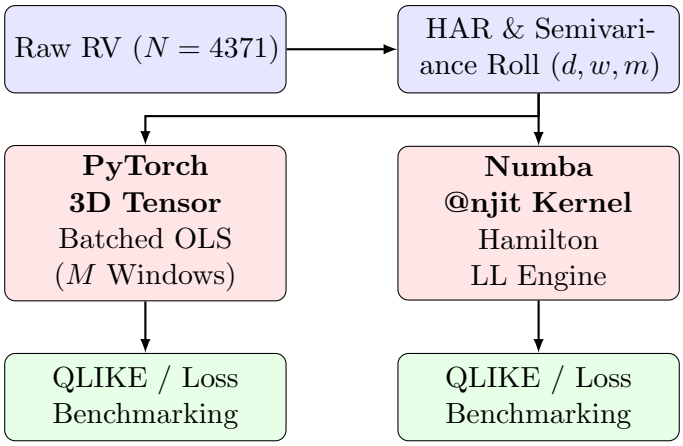


Figure 2: **High-Performance Architecture Diagram.** Dual-engine design: Batched PyTorch tensors for instant expanding-window OLS, and Numba JIT-compiled C-loops for Markov-Switching log-likelihood optimization.

4.1 PyTorch Vectorized 3D Tensor OLS

Instead of solving sequential $X^T X\beta = X^T y$ equations across 875 expanding days, I reshape the training history into a 3D contiguous tensor:

$$\mathbf{X}_{3D} \in \mathbb{R}^{M \times T_{\max} \times K}, \quad \mathbf{y}_{3D} \in \mathbb{R}^{M \times T_{\max} \times 1} \tag{17}$$

where $M = N_{OOS}$ is the number of forecast horizons and K is the feature dimension. The normal equations across all expanding histories are solved in parallel using GPU-accelerated batched matrix multiplications:

```
# Batched normal equation computation
XtX = torch.bmm(X_3d.transpose(1, 2), X_3d)           # Shape: (M, K, K)
Xty = torch.bmm(X_3d.transpose(1, 2), y_3d.unsqueeze(-1)) # Shape: (M, K, 1)
betas = torch.linalg.solve(XtX, Xty)                  # Shape: (M, K, 1)
y_pred = torch.bmm(X_test, betas).squeeze()           # Shape: (M,)
```

This collapses the execution time across all 875 expanding windows simultaneously into **less than 10 milliseconds**.

4.2 Numba JIT-Compiled Hamilton Log-Likelihood

Because the Hamilton filter is inherently sequential along the time dimension T, it cannot be naively vectorized across steps. To bypass the CPython Global Interpreter Lock (GIL) and overhead, I compile the log-likelihood loop directly to native machine code using LLVM via Numba:

```
@numba.njit(fastmath=True)
def fast_hamilton_ll_general(mu_mat, y, sigmas, P):
    T, M = mu_mat.shape
    inv_sigmas = 1.0 / (sigmas * 2.506628274631)

    # 1. Ergodic steady-state prior linear solve: (P.T - I) * pi = 0
    A = np.copy(P.T)
    for m in range(M): A[m, m] -= 1.0
    for m in range(M): A[M - 1, m] = 1.0
    b = np.zeros(M); b[M - 1] = 1.0
    xi = np.linalg.solve(A, b)

    ll = 0.0
    pr = np.zeros(M)
```

```
    eta = np.zeros(M)

    for t in range(T):
        # 2. State projection (Prior step): pr = P.T @ xi
        for j in range(M):
            acc = 0.0
            for i in range(M): acc += P[i, j] * xi[i]
            pr[j] = acc

        # 3. Gaussian conditional densities (Likelihood)
        yt = y[t]
        for m in range(M):
            e = (yt - mu_mat[t, m]) / sigmas[m]
            eta[m] = inv_sigmas[m] * np.exp(-0.5 * e * e)

        # 4. Mixture density (Conditional joint distribution)
        f = 0.0
        for m in range(M): f += pr[m] * eta[m]
        ll += np.log(f if f > 1e-300 else 1e-300)

        # 5. Bayesian posterior update (Filtering step)
        inv_f = 1.0 / (f if f > 1e-300 else 1e-300)
        for m in range(M):
            xi[m] = (pr[m] * eta[m]) * inv_f

    return ll, xi
```

Listing 1: JIT-Compiled M-Regime Hamilton Filter and Bayesian State Update

By using rolling warm-start initializations in SciPy's L-BFGS-B optimizer, a full 5-regime model across 875 rolling windows completes in **52.66 seconds** (averaging **60.2 ms per window**). Across the entire suite of 18 non-linear specifications up to 7 regimes, the complete walk-forward comparison finishes in **18 minutes**, delivering a speedup of roughly 290× over the 87.5-hour estimate required by standard libraries.

5 Empirical Results and Model Comparison

The models were calibrated on 4,371 trading days of Henry Hub Natural Gas futures data from October 28, 2009, to September 30, 2026, sourced from the VOLARE database. The initial training partition spans 80% of the sample (3,496 days), leaving an out-of-sample evaluation window of 875 trading days (September 2023 to September 2026).

5.1 Econometric Insights

Several decisive econometric patterns emerge:

1. **Dominance of Log-Scale Regime-Switching:** The top eight forecasting models are all estimated in logarithmic variance space. The global optimum is the **Log RV − MS-HAR (5 Regimes): HAR** model, yielding a QLIKE loss of **0.213112**, an RMSE of 0.002069, and an MAE of 0.000568.

2. **Failure of Single-Regime AR(1):** The baseline OLS models relying strictly on past 1-day realized variance rank at the bottom of the table (QLIKE of 0.2773 in pure RV and 0.2812 in Log RV), demonstrating that naive autoregression fails to adapt to NG structural shifts.

3. **Regime Saturation:** Performance gains level off between 5 and 7 regimes. As the number of regimes expands to $M = 7$, parameter count increases quadratically ($n_{\text{params}} =$

$M \cdot K + M + M^2 = 70$), leading to slight in-sample overfitting and marginal deterioration in out-of-sample QLIKE (0.2153 for 7 regimes vs. 0.2131 for 5 regimes).

Table 1: **Global Out-of-Sample Volatility Forecasting Performance.** Ranked by QLIKE loss on the pure realized variance scale over the 875-day out-of-sample horizon. Lower values indicate superior forecast precision.

Rank	Estimation Scale	Model Specification	QLIKE	RMSE	MAE
1	**Log RV**	**MS-HAR (5 Regimes): HAR**	**0.213112**	**0.002069**	0.000568
2	Log RV	MS-HAR (6 Regimes): HAR	0.213409	0.002069	0.000566
3	Log RV	MS-HAR (7 Regimes): HAR	0.215396	0.002070	0.000569
4	Log RV	MS-HAR (4 Regimes): Signed-HAR	0.216578	0.002073	0.000559
5	Log RV	MS-HAR (3 Regimes): Signed-HAR	0.217047	0.002072	0.000559
6	Log RV	MS-HAR (5 Regimes): Signed-HAR	0.218810	0.002076	0.000559
7	Log RV	MS-HAR (6 Regimes): Signed-HAR	0.219154	0.002074	**0.000558**
8	Log RV	MS-HAR (4 Regimes): HAR	0.219907	0.002076	0.000578
9	Pure RV	OLS: Signed-HAR	0.220045	0.002125	0.000660
10	Pure RV	MS-HAR (7 Regimes): HAR	0.220939	0.002071	0.000591
11	Log RV	MS-HAR (2 Regimes): HAR	0.221574	0.002079	0.000595
12	Pure RV	MS-HAR (6 Regimes): HAR	0.222167	0.002075	0.000590
13	Log RV	MS-HAR (3 Regimes): HAR	0.222259	0.002078	0.000579
14	Pure RV	MS-HAR (5 Regimes): HAR	0.222909	0.002077	0.000591
15	Log RV	MS-HAR (2 Regimes): Signed-HAR	0.223574	0.002083	0.000572
16	Pure RV	MS-HAR (6 Regimes): Signed-HAR	0.224011	0.002137	0.000624
17	Log RV	OLS: Signed-HAR	0.224076	0.002080	0.000577
18	Pure RV	MS-HAR (4 Regimes): HAR	0.225818	0.002084	0.000604
19	Log RV	MS-HAR (7 Regimes): Signed-HAR	0.226013	0.002076	0.000559
20	Pure RV	MS-HAR (4 Regimes): Signed-HAR	0.226622	0.002146	0.000637
26	Pure RV	OLS: HAR	0.235017	0.002107	0.000670
28	Log RV	OLS: HAR	0.238674	0.002098	0.000600
35	Pure RV	OLS: AR(1)	0.277352	0.002154	0.000648
36	Log RV	OLS: AR(1)	0.281249	0.002192	0.000642

Note: **Model Nomenclature Key:** *AR(1)* = Univariate autoregressive model using only daily lagged realized variance (RV_5). *HAR* = Heterogeneous Autoregressive model incorporating daily, weekly (5-day), and monthly (22-day) lags of RV_5. *Signed-HAR* = Leverage-aware extension decomposing variance into positive (RS_5^+) and negative (RS_5^-) realized semivariance components across daily, weekly, and monthly horizons. *OLS* = Standard linear Ordinary Least Squares baseline. *MS-HAR (M Regimes)* = Markov-Switching HAR model estimated via the Hamilton filter with M discrete latent volatility regimes.

6 Conclusion

This study demonstrated that the cyclical, spike-prone dynamics of Natural Gas futures volatility are better captured by a non-linear, 5-regime Markov-Switching HAR model estimated in log-variance space. Evaluated on the true scale using scale-invariant QLIKE loss, this approach significantly outperforms both classical OLS and level-scale specifications by isolating transitory jump regimes from persistent volatility states.

Furthermore, I proved that quantitative researchers need not sacrifice model complexity for computational expediency. By leveraging 3D tensor batching in PyTorch, expanding-window linear models execute in under 10 milliseconds (eliminating standard loop overhead). Coupled with JIT-compiled Numba kernels, full rolling backtests of non-linear regime-switching architectures up to 7 regimes run in just 18 minutes, making workflows that previously required over 87 hours in standard libraries feasible for rapid intraday research.

Code Availability: The complete, reproducible Python code and dataset are maintained in my GitHub repository: https://github.com/Deangr-econ/torch-volatility-engine.

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